

Canine and human factors related to dog bite injuries

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Objective—To identify physical traits of biting dogs and characteristics of injured persons and dog owners associated with bite situations for use in public health prevention activities.

Design—Retrospective cohort study.

Study Population—Licensed dog and dog bite report data from June 30, 2002, to July 1, 2003, that were obtained from Animal Control Services of Multnomah County, Oregon.

Procedures—To determine the canine and human factors associated with dog bite injuries, the number of bites, dog and injured person characteristics, and the overall canine population were evaluated. Dog owner characteristics at the block group level were defined by use of geographic information system software through 2000 census information based on place of residence.

Results—During the study period, 636 dog bites were reported to Animal Control Services, and 47,526 dogs were licensed in Multnomah County. Risk factors associated with biting dogs included breed (terrier, working, herding, and nonsporting breeds), being a sexually intact male, and purebred status. Male children aged 5 to 9 years had the highest rate of injury (178 bites/100,000 children). Biting dogs were more likely than nonbiting dogs to live in neighborhoods where the residents' median incomes were less than the county median income value (\$41,278).

Conclusions and Clinical Relevance—Dog bites continue to be a source of preventable injury. Prevention programs should target owners of sexually intact male and purebred dogs and owners who live in lower income neighborhoods. (*J Am Vet Med Assoc* 2008;232:542–546)

Dog bites are an important public health problem in the United States, and the number of bites reported each year appears to be increasing. In 1986, the National Health Interview Survey determined that 585,000 injuries requiring medical attention or resulting in restricted activity were associated with dog bites; this estimate placed dog bites among the top 12 causes of nonfatal injury in the United States.¹ Nine years later, the CDC estimated that 4.7 million dog bites occurred in the United States, of which approximately 800,000 required medical attention.² In 2001, an estimated 370,000 persons (130 bites/100,000 humans) sustained bites that were severe enough to require treatment in US emergency departments, with medical costs estimated at \$102.4 million.^{3,4} In that same year, 5,892 people were hospitalized because of dog bite injuries.³ In the United States, the mean number of people who die as a result of dog bite injury each year is 18.⁵

A survey conducted in 2001 by the American Pet Products Manufacturers Association estimated that 68 million dogs are owned in the United States; 40% of households have at least 1 dog, and 37% of those have more than 1 dog.⁶ The high number of dogs living in homes as pets results in the potential for large numbers of dog bite-related injuries. Factors that determine whether a dog-human interaction will result in a bite are complex and involve characteristics of the dog, the injured person,

ABBREVIATIONS	
MCAC	Multnomah County Animal Control
AKC	American Kennel Club

the owner, and the dog's environment. The purpose of the study reported here was to identify physical traits of biting dogs and characteristics of injured persons and dog owners associated with bite situations for use in public health prevention activities. The intent was to identify opportunities for bite prevention through the assessment of factors associated directly and indirectly with biting dogs.

Materials and Methods

Study population and data collection—By law, dog bite injuries are a reportable condition in Oregon.⁷ In Multnomah County, Oregon's most populous county with 677,813 residents,⁸ health-care providers, veterinarians, animal control personnel, law enforcement officials, and animal owners are required to report dog bites to MCAC within 1 working day.⁷ Presently, information on the physical traits including the AKC breed category, sex and neuter status (ie, neutered or sexually intact male and spayed or sexually intact female), and breed status (ie, mixed, pure, or unknown) is included in these records. Information regarding the bite situation is recorded, including relationship between the dog and the person injured, location at which the bite took place, date and time (ie, season), bite severity, and who reported the bite. The name, address, age, and sex of the injured person and the name and address of the dog owner are also required. Yet the information available is highly variable and often subjective, depending

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on the animal control officer documenting the information and the nature of the bite situation.

We evaluated MCAC reports of dog bites that occurred from June 30, 2002, to July 1, 2003. Dogs licensed in Multnomah County during this same period were used to describe the dog population for the county (denominator values) and calculate rates. Biting dogs included in the licensed dog database were included only once in the canine cohort. Nonlicensed dogs reported as biting were included with nonbiting licensed dogs in the total counts for the canine cohort. Human population denominator counts used to calculate dog bite rates were based on census data obtained from the US Census Bureau summary files and Portland State University Population Center.^{8,9}

Dog owner factors—Specific owner characteristics such as age, sex, or income level were not available from the bite reports or the licensed dog data; therefore, we assigned dog owner factors on the basis of characteristics of the neighborhood (defined by the census block group) in which the dog lived. Owner information for percentage sex by age, percentage nonwhite race, and percentage without a high school diploma (education level) was collected; median household income as well as the population per square mile (population density) of the block group in which the owner lived was recorded. Census variables were selected to approximate the social and socioeconomic conditions in which the dog (biting and nonbiting) lived. The dog owners' addresses were geocoded with Oregon Department of Human Services street data and geographic information system software.⁴ Initial coding was run with the criteria for an 80% match on the address and manually confirmed through the Portland Metropolitan Area Thomas Guide.¹⁰

Analysis—To determine factors associated with biting, rates of biting dogs were compared with all dogs in the county canine cohort by breed (sporting dogs as the referent category), sex-neuter status (spayed female as the referent category), and breed status (mixed breed as the referent category). Breed categories were used instead of specific breeds for calculating risk ratios because of the subjective nature of the breed listed on the bite reports and license records. A Pearson χ^2 test was used to assess independence in proportion of children (< 18 years old) bitten versus adults ($\geq$ 18 years old) and proportion of males versus females. Differences in rates and proportions were considered significant at a value of $P \leq 0.05$ or if 95% confidence intervals did not span 1. Characteristics of the dog bite situations were evaluated qualitatively by the highest reported proportions.

To evaluate dog neighborhood factors, multivariate regression analysis was performed, with biting as the outcome of interest matched on the breed category and by controlling for the sex of the dog. Dog neighborhood characteristics were derived from the census block group level; therefore, dogs (either biting or nonbiting) that lived in the same block group would have identical information. To account for this possible correlation effect, dog neighborhood characteristics were modeled with the use of generalized estimating equations logistic regression.^{11,b} Because of the significant association between breed category or sex-neuter status and the likelihood to bite identified with the cohort analysis and previous studies,^{5,12-19} breed category was frequency matched between biting and nonbiting dogs and sex-neuter status was controlled for in the final multivariate regression model.

The 7 broad AKC breed categories were used to describe all dogs in the study. The number of nonbiting dogs randomly selected from each category was dictated by the frequency of breed category distributions in the biting dog data set. Thirty nonbiting dogs were selected for each biting dog on the basis of these categories and proportions. The 486 biting dogs with valid census data were joined with the 14,580 breed category frequency-matched nonbiting dogs, thereby providing 15,066 dogs for the dog neighborhood analysis. The nonbiting dog data set contained only those breed categories represented in the biting dog data set and only dogs with a valid geocoded address.

Dog neighborhood characteristics, breed category, and sex-neuter status were evaluated individually for an association with biting outcome. All variables were placed in the multivariate regression model. The variable with the highest Wald statistic for type 3 generalized estimating equations P value was removed. The final model contained dog neighborhood factors with values of $P \leq 0.05$.

Results

Number of reported dog bites and characteristics of the dog population—During the 1-year study period, 636 dog bites were reported to MCAC, and 47,526 licensed dogs resided in Multnomah County. Characteristics of biting dogs were compared with those of all 47,850 dogs identified as residing in Multnomah County (Table 1). Of the 636 biting dogs, 312 (49%) had a license number on file with MCAC. Each dog's breed category, sex-neuter status, and purebred status were all associated ($P < 0.01$) with biting. Terrier, working, herding, and nonsporting breeds were more likely to bite than sporting breeds, whereas hounds, non-AKC breeds, and toy dogs were not significantly associated with biting. Bite reports were highest among sexually intact male dogs (risk ratio, 18.6; 95% confidence interval, 13.9 to 24.7) and purebred dogs (risk ratio, 3.8; 95% confidence interval, 2.9 to 5.0).

Characteristics of the dog bite situations and persons bitten by dogs—Characteristics of the 636 dog bite situations, including the relationship between the dog and the injured person, location at which the bite took place, the season, severity of the bite, and who reported the bite, were assessed (Table 2). The largest proportion (230/636 persons; 36%) of bite victims did not know the dogs that bit them. Yet among 41 children < 5 years old, 19 (46%) were bitten by the family dog. A higher proportion of bites took place in the dog's household (223/636 persons; 35%) and in the summer months (215/636 persons; 34%). Half of the injured persons sought medical care; health-care providers reported 43% (275/636 persons) of the bite events to MCAC.

During the study period, the reported dog bite rate in Multnomah County, Oregon, was 93 bites/100,000 persons (Table 3). There was no significant ($P = 0.4$) difference in bite rates between males and females. Boys aged 5 to 9 years had the highest rate of dog bite injuries (178 bites/100,000 children), which was significantly ($P = 0.01$) higher than the rate among other male age categories. Girls aged 5 to 9 years also had a high rate of dog bite injuries (125 bites/100,000 children), and this rate was not significantly ($P = 0.2$) different from the rate for boys aged 5 to 9 years. Boys < 18 years

Table 1—Breed and sex characteristics of 47,850 dogs, of which 636 were reported to have injured a person via biting in Multnomah County, Oregon, from June 30, 2002, to July 1, 2003.

Variable	No. of dogs in cohort	No. of reported biting dogs	Incidence bite rate (per 1,000 licensed dogs)	Risk ratio (95% confidence interval)
Breed category*				
Sporting	14,243	90	6.3	Referent
Terrier	4,798	114	23.8	3.8 (2.9–5.0)
Working	5,448	126	23.1	3.6 (2.8–4.8)
Herding	8,890	128	14.4	2.3 (1.7–3.0)
Nonsporting	5,262	66	12.5	2.0 (1.5–2.7)
Hound	3,881	28	9.8	1.1 (0.8–1.7)
Non-AKC	303	2	6.6	1.0 (0.3–4.2)
Toy	4,684	16	3.4	0.54 (0.03–0.9)
Unknown†	341	66		
Sex				
Spayed female	21,584	64	3.0	Referent
Sexually intact male	3,157	174	55.1	18.6 (13.9–24.7)
Sexually intact female	2,028	63	31.1	10.5 (7.4–14.8)
Neutered male	19,991	153	7.7	2.6 (1.9–3.5)
Unknown†	1,090	182		
Breed				
Mixed	17,326	187	10.8	Referent
Purebred	6,189	382	61.7	3.8 (2.9–5.0)
Unknown†	24,335	67		

*AKC breed categories. †Status not listed on bite report.

Table 2—Characteristics of 636 dog bite situations reported in Multnomah County, Oregon, from June 30, 2002, to July 1, 2003.

Characteristic	No. of bite reports (%)
Relationship between dog and victim	
No relationship	230 (36.2)
Family member	126 (19.8)
Neighbor	119 (18.7)
Unknown	86 (13.5)
Friend of the dog's family	75 (11.8)
Location of bite attack	
Dog's home or yard	223 (35.1)
Neutral*	149 (23.4)
Household of dog and victim†	108 (17.0)
Household of victim‡	64 (10.1)
Unknown	45 (7.1)
Place of employment§	24 (3.8)
Neighbor's property	23 (3.6)
Season	
Summer (Jun, Jul, Aug)	215 (33.8)
Spring (Mar, Apr, May)	158 (24.8)
Fall (Sept, Oct, Nov)	153 (24.1)
Winter (Dec, Jan, Feb)	105 (16.5)
Bite severity	
Severe—medical attention sought	321 (50.5)
Moderate—no medical attention sought but skin broken	206 (32.4)
Unknown	65 (10.2)
Mild—skin not broken	36 (5.7)
Person reporting bite	
Healthcare provider	275 (43.2)
Victim or guardian	230 (36.2)
Other	81 (12.7)
Unknown	26 (4.1)
Dog owner	24 (3.8)

*Neither the dog nor victim's environment (eg, sidewalk or park). †Household member bitten within residence of both dog and victim. ‡Victim's home or yard, not the dog's environment. §Place of employment such as a veterinary clinic, grooming facility, or MCAC. ||Biting dog owned by neighbor; bite situation occurred in the environment of the neighbor's dog.

Table 3—Incidence rates by age and sex of victim among 636 dog bite situations reported in Multnomah County, Oregon, from June 30, 2002, to July 1, 2003.

Sex of victim	Age (y)	Total population	No. of bite reports*	Incidence bite rate (per 100,000 persons)
Male	All ages	341,154	263	77
	0–4	24,054	21	87
	5–9	21,929	39	178
	10–17	34,082	34	100
	≥ 18 (adult)	261,089	169	65
Female	All ages	344,799	251	73
	0–4	22,952	20	87
	5–9	20,845	26	125
	10–17	32,335	15	46
	≥ 18 (adult)	268,667	190	71

*Sex and age of victim were identified in 514 bite reports.

old were more likely ($P < 0.01$) to be bitten than adult men. Among females, girls < 18 years old were not more likely ($P = 0.4$) to be bitten than adult women.

Characteristics of the dog owner—Dogs living in census block groups that had incomes less than the county median (\$41,278) were 1.5 times as likely to be reported as a biting dog than reported as a nonbiting dog (95% confidence interval, 1.3 to 1.9; $P < 0.01$) after matching on breed category and controlling for sex. The other dog owner factors examined, including the population density, percentage sex by age, percentage nonwhite race, and percentage without a high school diploma (education level), were not associated ($P > 0.05$) with a reported biting dog.

Discussion

During the period of study, a dog bit 93 of every 100,000 Multnomah County residents, as determined

from the reports to MCAC. Another study²⁰ evaluating bites reported to Animal Control Services in Kansas City, Mo, revealed a similar annual incidence of dog bites (92/100,000 members of the population). Emergency department data indicate that the national rates may be higher, with 130 bite injuries treated in emergency departments/100,000 members of the population.³ This would suggest that not all emergency department-treated bites are reported to animal control services.² In a dog bite epidemiologic review by Overall and Love,¹³ it was estimated that only 17% (253/1,489) of dog bites are reported to any authority.^{2,21} The true magnitude of the problem may be difficult to quantify with existing surveillance systems, yet continued research into dog bite situations will provide a valuable foundation for prevention efforts.

Similar to findings in previous studies,^{12–19} results of the present study indicated that the breed of dog played a role in whether a dog bit a person. Terriers including pit bull-type dogs, working dogs such as Rottweilers, and herding breeds such as German Shepherd Dogs have been bred to hunt vermin, protect property, and work livestock.²² In situations where they are not controlled, these dogs could revert to instinctual behaviors.^{14,15,23–25} In addition, bites from these breeds can result in more serious injury because of their size and strength.^{15,22}

Bite situations in Multnomah County are very similar to those that occur in the rest of the United States; most of the dog bite injuries reported nationally occur in the dog or victim's home, and the family dog is listed as the primary source of the bite.^{4,19,26} Children are most likely to be injured in this manner, and bites occur most frequently when the child is within the dog's territory.¹⁸ The association between bites and the dog's home environment could be explained by the need to express protective, possessive, or fear-induced aggression.^{13–15} Children, in particular, may not be able to discern between a dog that feels threatened and one that is playing.¹⁵ Dogs may be particularly protective of toys, food, and their yard.^{27,28} Given that 1 of every 5 households in Multnomah County has a licensed dog (47,526 licensed dogs and 272,098 households), there are many opportunities for dogs to bite people.⁸ Children are particularly vulnerable because of their small physical size and lack of experience to handle or defend themselves against an aggressive dog.^{16,25,26}

Innate tendencies dictated by breed, sex-neuter status, and size play a role in the potential of a dog to bite, but owners are ultimately responsible for their dogs' actions.^{15,29} An owner may want a dog with aggressive tendencies for protection purposes.^{14,15} Pit bull-type dogs, Rottweilers, and German Shepherd Dogs have large physical presences as well as reputations for aggressive behaviors.^{14,27} The personality and living environment of the owner may dictate not only the degree of socialization (and therefore the bite potential of the dog) but also the breed of dog they are likely to acquire.²⁹

The present study was unique in that it evaluated the dog owner's characteristics, as described by census or neighborhood data, as factors in dog bite incidents. Results of previous studies^{12,13,18,24} have indicated a strong association between both the breed of the dog and sex-neuter status and whether a dog bites. In our study, block group income levels appear to be a possible factor in whether a dog bites, irrespective of the dog's breed or sex. Dogs living in a lower income neighborhood may not be afforded the necessary training or supervision needed to minimize a high bite-risk situation. In another study²⁹ examining dog bite injuries in St Louis, Mo, bite injuries

occurring in low income areas were attributed to large numbers of children playing outdoors, few homes with adequate fencing, poor dog control, and a high proportion of large-breed dogs owned for protective purposes.

Dog bite studies, including that of this report, have inherent limitations. Besides the known underreporting of dog bites, breed-specific population estimates are lacking.^{14,30} Dog license data can only provide an estimate of breed-specific population denominators. In 1987, 43,650 dogs were licensed in Multnomah County, but a county-wide survey placed the true number of owned dogs at 101,794.³¹ Breed assessment is often subjective^{21,29}; license forms and bite reports rely on owners and injured persons to describe the breed. Bites from large-breed dogs, especially pit bull-type dogs, Rottweilers, and German Shepherd Dogs, are more likely to be reported and result in medical care, which would overrepresent those breeds among biting dogs—in other words, creating reporting bias.^{14–17} If owners of these particular breeds were also less likely to license their dogs, this would further skew the breed-specific bite rates.¹³ Analysis of data in the present study was unable to answer the question of whether certain breeds were more likely to be owned in lower income areas because of the breed category frequency-matched data used.

The human and veterinary medical communities and animal control agencies need to work together to help foster healthy relationships between people and their pets, especially in low income neighborhoods.^{21,26,29} Pediatricians in particular can play a vital role in injury-prevention counseling for children and parents (in both dog-owning and non-dog-owning families) during routine medical visits.³² Low-cost spay and neuter surgeries need to be easily accessible in the community, and education efforts should target dog owners through pet stores, groomers, boarding facilities, and other stakeholders such as postal workers and animal control agencies. Animal control agencies should be supported to maximize regulatory and educational activities. Nevertheless, owners are inevitably liable for the actions of their dogs and need to make every effort to minimize their dogs' bite potential through obedience training; neutering; and supervision, especially around children.

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- b. Statistical Analysis Software, version 9.1, SAS Institute, Cary, NC.

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Selected abstract for JAVMA readers from the American Journal of Veterinary Research

Inheritance of cataracts and primary lens luxation in Jack Russell Terriers

Anita M. Oberbauer et al

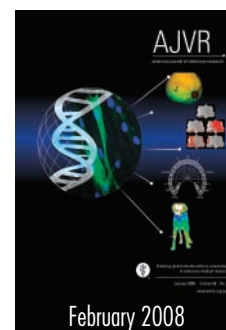
Objective—To characterize heritability and mode of inheritance of cataracts and primary lens luxation in Jack Russell Terriers.

Sample Population—872 Jack Russell Terriers from which buccal epithelial cells were collected and phenotypes for cataracts and lens luxation were determined and an additional 1,898 Jack Russell Terriers without phenotypic information used to complete pedigree relationships and that were included in the analyses.

Procedures—Narrow-sense heritabilities and genetic correlation for cataracts and lens luxation were modeled by use of threshold analysis, whereas complex segregation analysis was used to characterize mode of inheritance. For the analyses, dogs < 6 years old, unless confirmed as having cataracts or lens luxation, were classified as an unknown phenotype. The possible involvement of an HSF4 mutation in cataracts was determined by DNA sequencing.

Results—Cataracts and primary lens luxation were highly heritable and genetically correlated, and neither was controlled by a single gene. Cataracts were not associated with an HSF4 mutation.

Conclusions and Clinical Relevance—Analysis of the data indicated that concerted selection against both cataracts and primary lens luxation when choosing breeding animals can be used to improve ocular health in Jack Russell Terriers. (*Am J Vet Res* 2008;69:222–227)



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